

Systems of Equations

$$A\mathbf{x} = \mathbf{b}$$

$$\begin{aligned}
 a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1 \\
 a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= b_2 \\
 &\vdots \\
 a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n &= b_m,
 \end{aligned}$$

$$\mathbf{Ax} = \mathbf{b}$$

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}.$$

Gauss Elimination Method

- (1) Swap one equation for another.
- (2) Add or subtract a multiple of one equation from another.
- (3) Multiply an equation by a nonzero constant.

System of Equations

$$\begin{aligned}x + 2y - z &= 3 \\2x + y - 2z &= 3 \\-3x + y + z &= -6.\end{aligned}$$

Tableau form

$$\left[\begin{array}{ccc|c} 1 & 2 & -1 & 3 \\ 2 & 1 & -2 & 3 \\ -3 & 1 & 1 & -6 \end{array} \right].$$

a_{ii} 's = pivots

$$\left[\begin{array}{ccc|c} 1 & 2 & -1 & 3 \\ 2 & 1 & -2 & 3 \\ -3 & 1 & 1 & -6 \end{array} \right] \xrightarrow{\substack{\text{subtract } 2 \times \text{row 1} \\ \text{from row 2}}} \left[\begin{array}{ccc|c} 1 & 2 & -1 & 3 \\ 0 & -3 & 0 & -3 \\ -3 & 1 & 1 & -6 \end{array} \right]$$

$$\xrightarrow{\substack{\text{subtract } -3 \times \text{row 1} \\ \text{from row 3}}} \left[\begin{array}{ccc|c} 1 & 2 & -1 & 3 \\ 0 & -3 & 0 & -3 \\ 0 & 7 & -2 & 3 \end{array} \right]$$

$$\left[\begin{array}{ccc|c} 1 & 2 & -1 & 3 \\ 0 & -3 & 0 & -3 \\ 0 & 7 & -2 & 3 \end{array} \right] \xrightarrow{\substack{\text{subtract } -\frac{7}{3} \times \text{row 2} \\ \text{from row 3}}} \left[\begin{array}{ccc|c} 1 & 2 & -1 & 3 \\ 0 & -3 & 0 & -3 \\ 0 & 0 & -2 & -4 \end{array} \right]$$

$$\begin{aligned}x + 2y - z &= 3 \\-3y &= -3 \\-2z &= -4,\end{aligned}$$

Operation counts:

$$\left[\begin{array}{cccc|c} a_{11} & a_{12} & \dots & a_{1n} & b_1 \\ a_{21} & a_{22} & \dots & a_{2n} & b_2 \\ \vdots & \vdots & \dots & \vdots & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nn} & b_n \end{array} \right]$$

$$m = \frac{a_{21}}{a_{11}}$$

$$\sum_{j=1}^{n-1} \sum_{i=1}^j 2(j+1) + 1 = \sum_{j=1}^{n-1} 2j(j+1) + j$$

$$= 2 \sum_{j=1}^{n-1} j^2 + 3 \sum_{j=1}^{n-1} j = 2 \frac{(n-1)n(2n-1)}{6} + 3 \frac{(n-1)n}{2}$$

$$= \frac{2}{3}n^3 + \frac{1}{2}n^2 - \frac{7}{6}n, \quad n \gg 1$$

$$\approx 2n^3/3 \text{ operations. } O(n^3)$$

$$\left[\begin{array}{ccccccccc} 0 & & & & & & & & \\ 2n+1 & & 0 & & & & & & \\ 2n+1 & 2(n-1)+1 & & 0 & & & & & \\ 2n+1 & 2(n-1)+1 & 2(n-2)+1 & & 0 & & & & \\ \vdots & \vdots & \vdots & \ddots & \ddots & & & & \\ \vdots & \vdots & \vdots & & & & & & \\ 2n+1 & 2(n-1)+1 & 2(n-2)+1 & \dots & 2(3)+1 & & 0 & & \\ 2n+1 & 2(n-1)+1 & 2(n-2)+1 & \dots & 2(3)+1 & 2(2)+1 & & 0 & \end{array} \right].$$

Back-substitution: $O(n^2)$.

Total no. of operations: $O(n^3)$

Failure of Gauss method: zero pivot ($a_{ii} = 0$) or *swamping*

Consider the system of equations

$$\begin{aligned}10^{-20}x_1 + x_2 &= 1 \\ x_1 + 2x_2 &= 4.\end{aligned}$$

$$\left[\begin{array}{cc|c} 10^{-20} & 1 & 1 \\ 1 & 2 & 4 \end{array} \right] \xrightarrow{\text{subtract } 10^{20} \times \text{row 1} \\ \text{from row 2}} \left[\begin{array}{cc|c} 10^{-20} & 1 & 1 \\ 0 & 2 - 10^{20} & 4 - 10^{20} \end{array} \right].$$

Computer version 2.

$$(2 - 10^{20})x_2 = 4 - 10^{20} \longrightarrow x_2 = \frac{4 - 10^{20}}{2 - 10^{20}},$$

$$\left[\begin{array}{cc|c} 1 & 2 & 4 \\ 10^{-20} & 1 & 1 \end{array} \right] \xrightarrow{\text{subtract } 10^{-20} \times \text{row 1} \\ \text{from row 2}} \longrightarrow \left[\begin{array}{cc|c} 1 & 2 & 4 \\ 0 & 1 - 2 \times 10^{-20} & 1 - 4 \times 10^{-20} \end{array} \right].$$

$1 - 2 \times 10^{-20}$ is stored as 1 and $1 - 4 \times 10^{-20}$ is stored as 1.

$$\begin{aligned}x_1 + 2x_2 &= 4 \\ x_2 &= 1,\end{aligned}$$

$$x_1 = 2 \text{ and } x_2 = 1.$$

$$[x_1, x_2] = \left[\frac{2 \times 10^{20}}{10^{20} - 2}, \frac{4 - 10^{20}}{2 - 10^{20}} \right] \approx [2, 1].$$

Computer version 1.

$$-10^{20}x_2 = -10^{20} \longrightarrow x_2 = 1.$$

$$[x_1, x_2] = [0, 1].$$

Partial Pivoting

$$|a_{p1}| \geq |a_{i1}|$$

for all $1 \leq i \leq n$, and exchange rows 1 and p .

$$x_1 - x_2 + 3x_3 = -3$$

$$-x_1 - 2x_3 = 1$$

$$2x_1 + 2x_2 + 4x_3 = 0.$$

$$\left[\begin{array}{ccc|c} 1 & -1 & 3 & -3 \\ -1 & 0 & -2 & 1 \\ 2 & 2 & 4 & 0 \end{array} \right].$$

$$\left[\begin{array}{ccc|c} 1 & -1 & 3 & -3 \\ -1 & 0 & -2 & 1 \\ 2 & 2 & 4 & 0 \end{array} \right] \xrightarrow{\text{exchange row 1 and row 3}} \left[\begin{array}{ccc|c} 2 & 2 & 4 & 0 \\ -1 & 0 & -2 & 1 \\ 1 & -1 & 3 & -3 \end{array} \right]$$

$$\xrightarrow{\text{subtract } -\frac{1}{2} \times \text{row 1 from row 2}} \left[\begin{array}{ccc|c} 2 & 2 & 4 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & -1 & 3 & -3 \end{array} \right]$$

$$\xrightarrow{\text{subtract } \frac{1}{2} \times \text{row 1 from row 3}} \left[\begin{array}{ccc|c} 2 & 2 & 4 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & -2 & 1 & -3 \end{array} \right].$$

$$\left[\begin{array}{ccc|c} 2 & 2 & 4 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & -2 & 1 & -3 \end{array} \right] \xrightarrow{\text{exchange row 2 and row 3}} \left[\begin{array}{ccc|c} 2 & 2 & 4 & 0 \\ 0 & -2 & 1 & -3 \\ 0 & 1 & 0 & 1 \end{array} \right]$$

$$\xrightarrow{\text{subtract } -\frac{1}{2} \times \text{row 2 from row 3}} \left[\begin{array}{ccc|c} 2 & 2 & 4 & 0 \\ 0 & -2 & 1 & -3 \\ 0 & 0 & \frac{1}{2} & -\frac{1}{2} \end{array} \right]$$

Resolves zero pivoting and swamping

LU decomposition

Equation of motion

$$m \frac{dv}{dt} = F(t)$$

Laplace's equation with a time-varying source

$$\nabla^2 V = \rho(t)$$

$$A_{ij}V_j = \rho_i, \quad (i = \text{site index})$$

$$L = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -3 & -\frac{7}{3} & 1 \end{bmatrix}.$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -3 & -\frac{7}{3} & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & -1 \\ 0 & -3 & 0 \\ 0 & 0 & -2 \end{bmatrix} = \begin{bmatrix} 1 & 2 & -1 \\ 2 & 1 & -2 \\ -3 & 1 & 1 \end{bmatrix} = A.$$

Decompose $A=LU$

An $m \times n$ matrix L is **lower triangular** if its entries satisfy $l_{ij} = 0$ for $i < j$. An $m \times n$ matrix U is **upper triangular** if its entries satisfy $u_{ij} = 0$ for $i > j$. $\square$

$$A = \begin{bmatrix} 1 & 2 & -1 \\ 2 & 1 & -2 \\ -3 & 1 & 1 \end{bmatrix}.$$

$$\begin{bmatrix} 1 & 2 & -1 \\ 2 & 1 & -2 \\ -3 & 1 & 1 \end{bmatrix} \xrightarrow{\substack{\text{subtract } 2 \times \text{row 1} \\ \text{from row 2}}} \begin{bmatrix} 1 & 2 & -1 \\ 0 & -3 & 0 \\ -3 & 1 & 1 \end{bmatrix}$$

$$\xrightarrow{\substack{\text{subtract } -3 \times \text{row 1} \\ \text{from row 3}}} \begin{bmatrix} 1 & 2 & -1 \\ 0 & -3 & 0 \\ 0 & 7 & -2 \end{bmatrix}$$

$$\xrightarrow{\substack{\text{subtract } -\frac{7}{3} \times \text{row 2} \\ \text{from row 3}}} \begin{bmatrix} 1 & 2 & -1 \\ 0 & -3 & 0 \\ 0 & 0 & -2 \end{bmatrix} = U.$$

Properties of the **L** matrix:

- (1) Let $L_{ij}(-c)$ denote the lower triangular matrix whose only nonzero entries are 1's on the main diagonal and $-c$ in the (i, j) position. Then $A \rightarrow L_{ij}(-c)A$ represents the row operation “subtracting c times row j from row i .”

$$\begin{aligned} A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} &\xrightarrow{L_{21}(-c)} \begin{bmatrix} 1 & 0 & 0 \\ -c & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \\ &= \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} - ca_{11} & a_{22} - ca_{12} & a_{23} - ca_{13} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \end{aligned}$$

- (2) $L_{ij}(-c)^{-1} = L_{ij}(c)$.
For example,

$$\begin{bmatrix} 1 & 0 & 0 \\ -c & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ c & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

(3)

$$\begin{bmatrix} 1 & & \\ c_1 & 1 & \\ & & 1 \end{bmatrix} \begin{bmatrix} 1 & & \\ & 1 & \\ c_2 & & 1 \end{bmatrix} \begin{bmatrix} 1 & & \\ & 1 & \\ & c_3 & 1 \end{bmatrix} = \begin{bmatrix} 1 & & \\ c_1 & 1 & \\ c_2 & c_3 & 1 \end{bmatrix}.$$

Demonstration of LU decomposition:

$$A = \begin{bmatrix} 1 & 2 & -1 \\ 2 & 1 & -2 \\ -3 & 1 & 1 \end{bmatrix}.$$

$L_{32}(7/3)$ $L_{31}(3)$ $L_{21}(-2)$

$$\begin{bmatrix} 1 & & \\ & 1 & \\ & \frac{7}{3} & 1 \end{bmatrix} \begin{bmatrix} 1 & & \\ & 1 & \\ 3 & & 1 \end{bmatrix} \begin{bmatrix} 1 & & \\ -2 & 1 & \\ & & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & -1 \\ 2 & 1 & -2 \\ -3 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 2 & -1 \\ 0 & -3 & 0 \\ 0 & 0 & -2 \end{bmatrix} = U$$

$L_{21}(2)$ $L_{31}(-3)$ $L_{32}(-7/3)$

$$A = \begin{bmatrix} 1 & & \\ 2 & 1 & \\ & & 1 \end{bmatrix} \begin{bmatrix} 1 & & \\ & 1 & \\ -3 & & 1 \end{bmatrix} \begin{bmatrix} 1 & & \\ & 1 & \\ & -\frac{7}{3} & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & -1 \\ 0 & -3 & 0 \\ 0 & 0 & -2 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & & \\ 2 & 1 & \\ -3 & -\frac{7}{3} & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & -1 \\ 0 & -3 & 0 \\ 0 & 0 & -2 \end{bmatrix} = LU.$$

$$Ax = b$$

$$\Rightarrow LUx = b.$$

(a) Solve $Lc = b$ for c .

(b) Solve $Ux = c$ for x .

Total no. of operations: $O(n^3)$

A **permutation matrix** is an $n \times n$ matrix consisting of all zeros, except for a single 1 in every row and column. $\square$

Permutation matrices:

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix},$$
$$\begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$$

Example:

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} = \begin{bmatrix} a & b & c \\ g & h & i \\ d & e & f \end{bmatrix}.$$

Memory aid:

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}.$$

PA=LU factorization

$$A = \begin{bmatrix} 2 & 1 & 5 \\ 4 & 4 & -4 \\ 1 & 3 & 1 \end{bmatrix}.$$

$$\begin{bmatrix} 2 & 1 & 5 \\ 4 & 4 & -4 \\ 1 & 3 & 1 \end{bmatrix} \xrightarrow{\substack{P = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \\ \text{exchange rows 1 and 2}}} \begin{bmatrix} 4 & 4 & -4 \\ 2 & 1 & 5 \\ 1 & 3 & 1 \end{bmatrix}$$

$$\xrightarrow{\substack{\text{subtract } \frac{1}{2} \times \text{row 1} \\ \text{from row 2}}} \begin{bmatrix} 4 & 4 & -4 \\ \frac{1}{2} & -1 & 7 \\ 1 & 3 & 1 \end{bmatrix} \xrightarrow{\substack{\text{subtract } \frac{1}{4} \times \text{row 1} \\ \text{from row 3}}} \begin{bmatrix} 4 & 4 & -4 \\ \frac{1}{2} & -1 & 7 \\ \frac{1}{4} & 2 & 2 \end{bmatrix},$$

$$\xrightarrow{\substack{P = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix} \\ \text{exchange rows 2 and 3}}} \begin{bmatrix} 4 & 4 & -4 \\ \frac{1}{4} & 2 & 2 \\ \frac{1}{2} & -1 & 7 \end{bmatrix}$$

$$\xrightarrow{\substack{\text{subtract } -\frac{1}{2} \times \text{row 2} \\ \text{from row 3}}} \begin{bmatrix} 4 & 4 & -4 \\ \frac{1}{4} & 2 & 2 \\ \frac{1}{2} & -\frac{1}{2} & 8 \end{bmatrix}.$$

$$\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 2 & 1 & 5 \\ 4 & 4 & -4 \\ 1 & 3 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ \frac{1}{4} & 1 & 0 \\ \frac{1}{2} & -\frac{1}{2} & 1 \end{bmatrix} \begin{bmatrix} 4 & 4 & -4 \\ 0 & 2 & 2 \\ 0 & 0 & 8 \end{bmatrix}$$

$P \qquad A \qquad L \qquad U$